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Smart Thermal System for Patient Safety Monitoring

Fire Detection — Algorithm Evaluation Report

MLX90640 Thermal Sensor 22 Scenes 5,943 Annotated Frames

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1 Overview

This report evaluates the fire detection algorithms of the Smart Thermal System (§4.4.4 of the engineering report) on the full annotated dataset. Three detector configurations are compared:

1. **Otsu Pipelined (original)** — the rule-based three-stage detector using Otsu segmentation plus morphological erosion/dilation, as originally specified.
2. **Otsu Hot-Pixel Bypass (improved)** — the same detector after a targeted fix that segments directly on an absolute temperature threshold and removes the erosion step (Section 6).
3. **Fire SVM** — the feature-based RBF support-vector classifier.

Every frame is reduced to a binary verdict — **Fire** (an alert of any level) or **Safe** — and scored with a confusion matrix and the four derived metrics (accuracy, precision, recall, F1). The central finding is that fire detection on this sensor is fundamentally limited by the *thermal signature* of the labelled fires, not only by algorithm design: the headline numbers must be read together with the data analysis in Section 7.

2 Dataset and Experimental Setup

2.1 Dataset

Recordings were collected at the Beer Sheva Mental Health Center using three MLX90640 sensors (32×24 px, 8 Hz). The evaluation uses **22 scenes** totalling **5,943 annotated frames** across all three channels. Fire is annotated as YOLO class 0; a frame is a fire *positive* if it carries at least one fire box.

Table 1: Fire-positive vs. fire-negative frame counts (all channels).

Scene group	Fire frames (Pos)	Safe frames (Neg)
`3pplfire`	42	213
`onepersonfire`	31	230
`setup2_fire_lighter`	5	301
Hot distractors (hairdryer, PC screen)	0	528
Other 17 scenes (people, motion)	0	4,393
Total	**78**	**5,865**

The dataset is **extremely imbalanced**: only 78 of 5,943 frames ($\approx 1.3\,\%$) contain fire, spread across just three scenes. Two scenes are deliberately retained as *hard negatives*: `2pplwithhairdryer` (a hot air jet) and `emptyroomwithpcscreen` (a warm monitor). These contain genuinely hot regions but no fire, and are the most demanding test of precision.

2.2 Evaluation Protocol

The two **Otsu** configurations are rule-based and require no training, so they are evaluated on the **full dataset** (all 5,943 frames). The **Fire SVM** is trainable and is evaluated on a held-out **70 / 30 test split** stratified per (scene, channel, label): 4,128 training frames (53 positive) and 1,815 test frames (25 positive). For a fair head-to-head, the improved Otsu detector is *also* reported on that same test split in Section 8. The detector's temporal state is reset at every scene boundary.

3 Metric Definitions

3.1 Confusion Matrix

Each frame yields one of four outcomes:

- **TP** (True Positive): fire present and correctly alarmed.
- **TN** (True Negative): no fire and system correctly silent.
- **FP** (False Positive): no fire but alarm raised — a *false alarm*.
- **FN** (False Negative): fire present but missed — a *missed fire*.

3.2 Derived Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Accuracy (Eq. 1) is the fraction of all frames classified correctly. Under 1.3 % positives it is **badly misleading**: a detector that never alarms scores 98.7 % accuracy while missing every fire. Accuracy is reported for completeness only and should not be used to rank detectors here.

Precision (Eq. 2) is the fraction of fire alarms that are genuine. Low precision means frequent false alarms, which cause *alert fatigue* and lead staff to disable the system — especially damaging in a psychiatric ward.

Recall (Eq. 3) is the fraction of real fires detected. It is the **primary safety metric**: a missed ignition is a direct, potentially life-threatening system failure.

F1-Score (Eq. 4) is the harmonic mean of precision and recall. It is the headline metric because accuracy is uninformative under this imbalance.

4 Algorithms

Otsu Pipelined (original) — rule-based, no training.
Variance gate → Otsu segmentation → 1× erosion + 2× dilation → hierarchical $T_{\max}$/area rules → temporal mass-gradient growth check.

Otsu Hot-Pixel Bypass — rule-based, no training.
Same pipeline, but segmentation thresholds directly at T_{ign} via connected components (pixel-count area, no erosion), preceded by dead-pixel despiking. See Section 6.

Fire SVM — classical ML, RBF support-vector classifier.
Otsu region proposal → 6 scalar blob features (T_{max}, T_{mean}, T_{std}, area, skewness, kurtosis) → standardisation → RBF SVM decision.

Decision thresholds (both Otsu variants): $T_{ign} = 45\,°C$, $T_{fire} = 60\,°C$, $A_{limit} = 23\,px$ ($\approx 3\,\%$ of the frame).

5 Results — Otsu Detector (full dataset)

5.1 Confusion Matrices

Otsu Pipelined (original)

	Pred: Fire	**Pred: Safe**
Truth: Fire (78 frames)	**TP** = 7	**FN** = 71
Truth: Safe (5865 frames)	**FP** = 3	**TN** = 5862

Otsu Hot-Pixel Bypass (improved)

	Pred: Fire	**Pred: Safe**
Truth: Fire (78 frames)	**TP** = 21	**FN** = 57
Truth: Safe (5865 frames)	**FP** = 31	**TN** = 5834

5.2 Per-Scene Breakdown

Table 2: Per-scene confusion counts on the three fire scenes and two hot distractors (full dataset). "Orig." = Otsu Pipelined; "Bypass" = Hot-Pixel Bypass.

	Original				**Bypass**			
Scene	TP	TN	FP	FN	TP	TN	FP	FN
`3pplfire`	7	213	0	35	20	213	0	22
`onepersonfire`	0	230	0	31	1	228	2	30
`setup2_fire_lighter`	0	301	0	5	0	299	2	5
`2pplwithhairdryer`	0	298	2	0	0	288	12	0
`emptyroomwithpcscreen`	0	228	0	0	0	224	4	0

5.3 Metric Summary

Table 3: Otsu detector — full-dataset metrics (5,943 frames).

Configuration	Acc	Prec	Rec	F1	TP	TN	FP	FN
Otsu Pipelined (original)	98.8%	70.0%	9.0%	**15.9%**	7	5862	3	71
Otsu Hot-Pixel Bypass	98.5%	40.4%	**26.9%**	**32.3%**	21	5834	31	57

6 The Hot-Pixel Bypass Fix

The original detector's recall of 9.0 % was traced to two interacting defects in the segmentation stage:

1. **Erosion deletes single-pixel fires.** At 32×24 resolution a real fire is frequently a *single* hot pixel. In `onepersonfire` one frame contains a genuine 72.5 °C pixel, yet the 1× erosion (3×3 kernel) removes any pixel lacking hot 8-connected neighbours, deleting the fire entirely before classification.
2. **Otsu's threshold is anchored by warm bodies.** With people in frame, Otsu selects a threshold around 25 °C, so the dominant "hot" blob is a person (≈ 39 °C) and the 72.5 °C fire pixel is never isolated as its own region.

The fix replaces Otsu+erosion in the alert path with direct thresholding at T_{ign} using connected-component labelling (*pixel-count* area, so a lone pixel has area 1, not 0), preceded by a despiking filter that neutralises isolated dead-pixel glitches (single pixels reading 800–935 °C). The hierarchical classifier already acted only on blobs with $T_{\max} \geq T_{\text{ign}}$, so this is equivalent in intent while recovering the small fires erosion discarded.

Effect. Recall improved **9.0** % → **26.9** % (3×, TP 7 → 21) and F1 **15.9** % → **32.3** %. Localisation quality on `3pplfire` rose from a mean IoU of 1.0 % to 17.3 %. The cost is precision: 70 % → 40.4 % (FP 3 → 31), because thresholding at 45 °C now also flags warm distractor pixels — most visibly the hairdryer jet (12 FP) and PC screen (4 FP). This precision/recall trade-off is examined in Sections 7 and 10.

7 The Recall Ceiling — A Data Limitation

The improved recall of 26.9 % is not an algorithmic shortfall; it is essentially the **physical ceiling** imposed by the labels. Analysing the 78 fire-positive frames:

Table 4: Peak-temperature analysis of the 78 fire-labelled frames.

Property	Value
Median peak temperature of a fire frame	36.5 °C
Minimum peak temperature	28.7 °C
Fire frames with any pixel > 45 °C (T_{ign})	21 / 78 (26.9 %)
Fire frames with any pixel > 60 °C (T_{fire})	9 / 78 (11.5 %)

The median fire frame peaks at only 36.5 °C — indistinguishable from a human body (33–37 °C)

— and **73 % of fire-labelled frames contain no pixel above the** 45 °**C ignition threshold**. The annotation boxes the fire *object* wherever it is visible (smouldering, distant, or sub-pixel), not only frames carrying a hot thermal signature. Consequently **no absolute-temperature detector can exceed** ≈ 27 % **recall** on this label set, and the bypass detector has now reached that ceiling. This is the single most important conclusion of the study: meaningful further gains require re-annotation, not algorithm tuning (Section 11).

8 Head-to-Head — Test Split

To compare the trainable SVM against the rule-based detector on identical data, both are evaluated on the same held-out test split (1,815 frames, 25 positive).

8.1 Confusion Matrices

Otsu Hot-Pixel Bypass

	Pred: Fire	**Pred: Safe**
Truth: Fire (25 frames)	**TP** = 5	**FN** = 20
Truth: Safe (1790 frames)	**FP** = 3	**TN** = 1787

Fire SVM (RBF)

	Pred: Fire	**Pred: Safe**
Truth: Fire (25 frames)	**TP** = 8	**FN** = 17
Truth: Safe (1790 frames)	**FP** = 0	**TN** = 1790

8.2 Metric Summary

Table 5: Test-split metrics (1,815 frames, 25 positive).

Detector	**Acc**	**Prec**	**Rec**	**F1**	**TP**	**TN**	**FP**	**FN**
Otsu Hot-Pixel Bypass	98.7%	62.5%	20.0%	**30.3%**	5	1787	3	20
Fire SVM (RBF)	99.1%	**100.0%**	**32.0%**	**48.5%**	8	1790	0	17

9 Consolidated Comparison

Table 6: Complete metric comparison across all three fire-detector configurations.

Detector	Eval set	Accuracy	Precision	Recall	F1
Otsu Pipelined (original)	Full (5,943)	98.8%	70.0%	9.0%	15.9%
Otsu Hot-Pixel Bypass	Full (5,943)	98.5%	40.4%	26.9%	32.3%
Otsu Hot-Pixel Bypass	Test (1,815)	98.7%	62.5%	20.0%	30.3%
Fire SVM (RBF)	Test (1,815)	99.1%	**100.0%**	**32.0%**	**48.5%**

10 Interpretation

Fire SVM is the strongest detector. At F1 = 48.5 % it leads every configuration, and it is the only detector with **zero false alarms** — crucially, it correctly stays silent on *both* hot distractors (hairdryer and PC screen). It achieves this by classifying on blob *shape* statistics (skewness, kurtosis, T_{std}, area) rather than temperature alone: a compact, sharply peaked fire signature is separable from a broad, smooth hot surface even when their peak temperatures coincide. On the common test split it beats the rule-based detector on every metric (+12 pp recall, +37.5 pp precision, +18.2 pp F1).

The Otsu bypass is a high-recall, low-precision gate. The fix tripled recall and is the right behaviour for a rule-based first stage, but a pure temperature threshold *cannot* separate a 50 °C fire pixel from a 50 °C hairdryer pixel — they are thermally identical. Its false alarms arrive through the immediate Ignition-Source path, which by design alarms instantly on any small hot blob and therefore bypasses the temporal growth check that would otherwise reject static heat sources.

Accuracy is meaningless here. All four rows in Table 6 report 98.5–99.1 % accuracy despite F1 ranging from 15.9 % to 48.5 %. This is the imbalance artefact warned of in Eq. 1: with 98.7 % of frames negative, even a near-blind detector looks "accurate". F1 and recall are the only honest rankings.

Recall is capped by the labels, not the model. Both detectors plateau near the 27 % ceiling of Section 7. The SVM slightly exceeds the naive > 45 °C threshold by learning warm-but-not-glowing fire textures, but the ≈ 73 % of sub-45 °C fire frames remain out of reach for any detector operating on this sensor.

11 Conclusions and Recommendations

11.1 Findings

1. **Fire SVM** is the recommended primary detector: F1 = 48.5 %, precision = 100 %, zero false alarms including on hot distractors.
2. The **hot-pixel bypass** fixed a genuine segmentation bug, tripling the rule-based detector's recall (9.0 $\rightarrow$ 26.9 %) and restoring usable localisation (IoU 1 $\rightarrow$ 17 %); it is best used as a high-recall pre-filter, not a standalone alarm.

3. **Accuracy is not a valid metric** under this 1.3 % imbalance; F1 and recall govern all conclusions.
4. Recall is bounded at ≈ 27 % by a **label/physics mismatch**: most fire-labelled frames carry no above-threshold thermal signature.

11.2 Recommended Actions

1. Re-annotate fire by thermal signature (highest leverage).
Decide whether "fire" denotes a *visible object* or a *detectable thermal hazard*. For a thermal-only safety system it must be the latter. Until the sub-45 °C frames are relabelled or excluded, the 27 % recall ceiling is immovable regardless of algorithm.

2. Add temporal confirmation to the ignition path.
Extending the mass-gradient persistence check to small IGNITION-SOURCE blobs (not only large POTENTIAL-FIRE blobs) should reject the static hairdryer/PC hot spots and recover much of the precision lost by the bypass, without sacrificing the recall gain.

3. Deploy the SVM with imbalance handling.
Retrain with class weighting or positive oversampling and add temporal/shape features to trade some of its perfect precision for higher recall, which is the priority safety metric.

4. Keep the despiking filter in preprocessing.
Dead-pixel glitches (800–935 °C) are a real false-alarm hazard for any $T_{\max}$ rule and must be neutralised before field deployment.